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FACULTY OF MATHEMATICS AND COMPUTER SCIENCE
DEPARTMENT OF COMPUTER SCIENCE

Information Systems

Lectures, tutorials and practical work

Second Year Computing

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Part One Course

Chapter 1

The company and its information system

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1.1 General information about the company

1.1.1 Definition

The company is part of society, a social body with legal, financial and social aspects. It is governed by its environment, just as it governs that environment because :

— It keeps pace with technical developments;
— It fits into the economic context;
— It adapts to the social system.

The company is organised around :

— Material resources: are the physical resources used by the company to manufacture goods or provide services.
— Human resources: staff resources in all categories.
— Methods (techniques): these are the processes carried out by the company's staff using material resources to carry out the company's task.

1.1.2 Objectives

The resources made available to the company are used to carry out its mission. This can be defined as the production of goods or services required to satisfy individual and collective needs.

Examples:

— Goods production company : Condor, LG, Sony; etc.
— Service providers: telephone operators, universities, etc.

1.1.3 Company classification

Two criteria are used to classify companies:

a) **Depending on the type of business** :

— *Commercial companies*: These buy goods, known as "merchandise", to resell them in the same condition as they were purchased, without undergoing any transformation.
— *Industrial companies*: Their purpose is to produce goods; they buy, process and sell them.
— *Service provider companies*: Their mission is to provide a service, i.e. to carry out certain useful tasks (e.g. universities, hospitals, etc.).

b) **According to the type of ownership of their assets :**

— Public companies: whose assets and liabilities consist entirely of public property.
— Private companies: whose assets belong to one person (sole proprietorship) or to several people (partnership).

1.1.4 Organisation of the company

Organising means :

— Define and assign tasks to each person
— Grouping tasks and individuals
— Defining a hierarchy
— Design a communication system between individuals and between departments
— Monitor system operation (manager).

The company's main activities, which contribute to the same objective, can be defined by as many functions. For example: production, marketing, etc.

When the activities relating to a function are large and varied, the company is structured into several *departments*.

Activity: the company appears in society as the place for action and to describe it: every company consumes inputs, implements techniques and resources, and produces outputs (see 1.1).

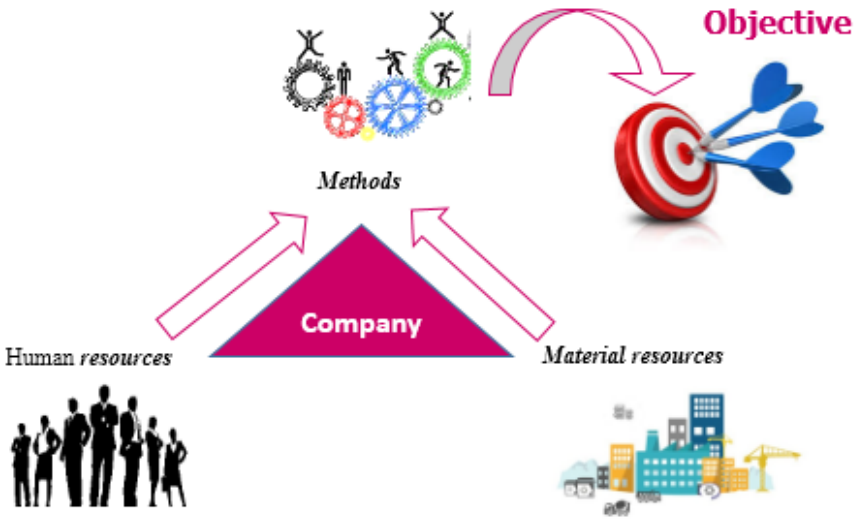


FIGURE 1.1: Business activity.

A department: is made up of a group of individuals, with material resources, performing all or part of a function.

Example: Our IT department is made up of several departments: schooling, diplomas, pedagogy, etc.

The organisation chart: is a diagram showing how the company is organised. It shows the functions and departments, the names of managers and the hierarchy (for more details, see TP1).

1.1.5 Company environment

The company maintains relationships with a number of partners:

— Other companies, suppliers, customers or subcontractors, service providers.
— Financial partners, banks or other companies.
— Government departments, taxes, social security contributions...

— Local authorities, town halls, departments, regions...

So the company is not isolated; it needs many partners. Figure 1.2 shows a diagram of the business environment.

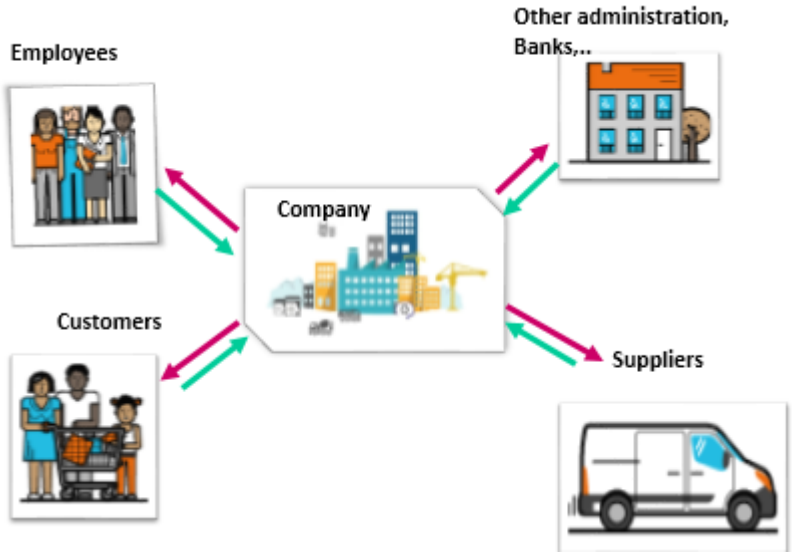


FIGURE 1.2: Company environment.

1.1.6 Systematic representation of the company

Definition o f a system :

A system can be defined as a set of dynamically interacting tangible or intangible elements organised to achieve a goal. To achieve this goal, the system takes account of its environment and regulates its operation by adapting to changes. The interaction between the system and its environment is made possible by information flows. These flows also circulate within the system, enabling it to analyse its own functioning (see Figure 1.3).

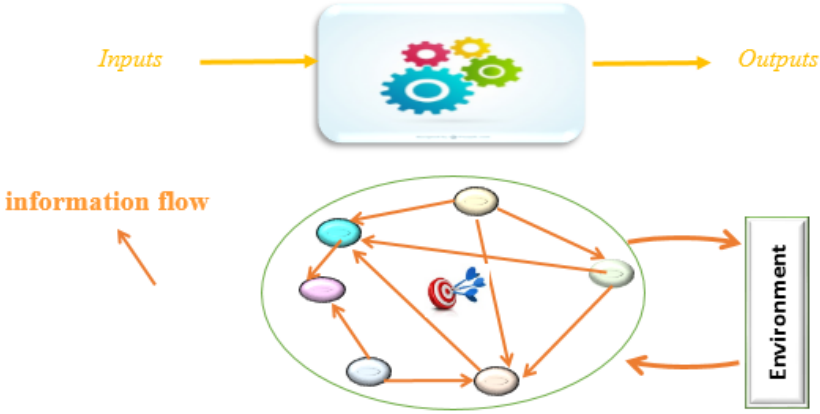


FIGURE 1.3: Definition of a system.

The company can be seen as a system:

— Tangible or intangible elements: material and human resources and applied technology.

— The company's environment: public authorities, customers, suppliers. The

company is a system made up of three sub-systems:

1. *Operating system (OS)*: This is responsible for carrying out all operations. It encompasses all the functions linked to the company's own activity: invoicing customers, paying employees, managing stocks.
2. *Decision-making system (DS)*: the process of steering (regulating and controlling) the SO by deciding how it should behave according to the objectives set. This system is made up of general management, commercial management, financial management, etc. It receives information from the SO on its status and reacts by taking decisions on the SO process by regulating flows (launching a new product range, changing prices, etc.).
3. *Information system :*
   The IS is the interface between the SO and the SD. It is used to ensure coupling and synchronisation between the two systems by exchanging and processing information. It is responsible for storing and processing information relating to the SO in order to make it available to the SD.

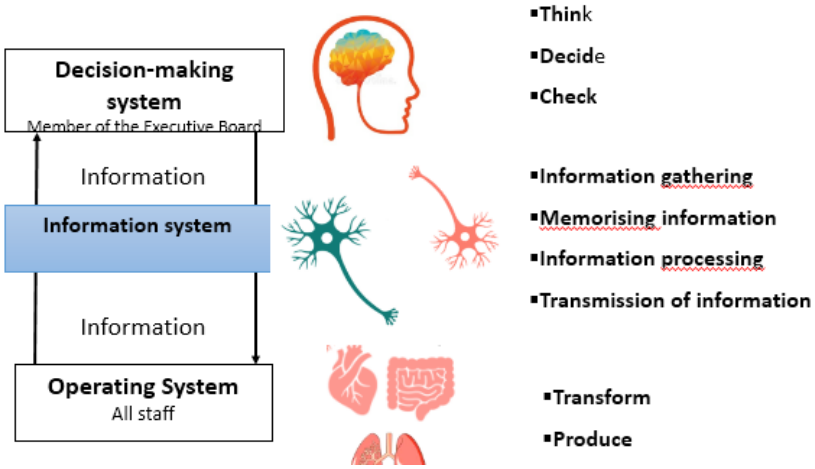


FIGURE 1.4: Systematic representation of the company.

1.2 Information System (IS)

1.2.1 Definition

An IS is an organised set of resources: hardware, software, personnel, data, procedures, etc. used to acquire, process and store information (in the form of data, text, images, sound, etc.) within and between organisations. Figure 1.5 shows the different functionalities of an IS.

The information system can therefore be defined as all the information flows circulating within the organisation, combined with the resources implemented to manage them.

The resources mobilised to manage the system are human and material. The information system is increasingly automated, meaning that the information is generated by machines (automata and computers). However, part of the system remains the responsibility of the company's human resources (data entry, decisions, interpretations).

Computers and humans use rules and procedures to process information. For example: mathematical models, algorithms, standards, regulations, administrative procedures.

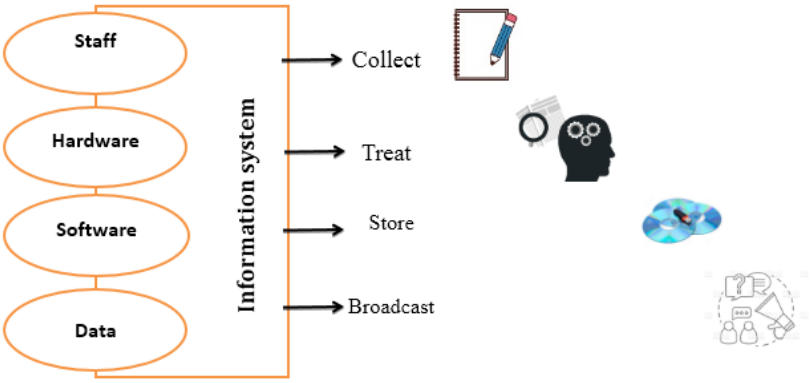


FIGURE 1.5: Definition of an IS.

1.2.2 Information system functions

1.2.2.1 Information gathering

To function, the system needs to be powered. The information comes from various sources:

— External sources: the information comes from the system's environment. This generally involves flows from the system's partners (customers, suppliers, government departments, etc.). Increasingly, companies need to listen to their environment in order to anticipate changes and adapt their operations. The development of means of communication (particularly the internet) means that information can be found more easily, but using it remains tricky (quality and reliability of information).

— Internal sources: the information system must be fed by flows generated by the various players in the system. These flows result from the system's activity: supplies, production, employee management, accounting, sales, etc.

The information deemed relevant to the organisation must be entered. This operation is generally costly, as it often requires human intervention. Many efforts have been made to automate information gathering (real-time systems, scanning, digitisation, content analysis robots, etc.).

1.2.2.2 Memorising information

Once the information has been captured, it needs to be preserved, i.e. stored in a durable and reliable way. Today, the preferred medium for storing information is the Internet.
made available by computer disks (magnetic or optical: hard disks, CDs, DVDs, etc.); however, paper is still a widely used medium in companies (storage of paper archives). The information stored on computers is either in the form of a file or organised to make it easier to use in the form of a database.

1.2.2.3 Information processing

To be usable, the information has to be processed. Once again, processing can be manual (which is less and less often the case) or automatic (carried out by computers). The main types of processing consist of searching for and extracting information, consolidating and comparing information, modifying and deleting information or producing new information by applying

calculations.

1.2.2.4 Dissemination of information

In order to be used, information must reach its recipient as quickly as possible. There are many ways of disseminating information: on paper, orally and, more and more often, using digital media which guarantee optimum transmission speed and the possibility of reaching as many people as possible. This is all the more true in the age of the Internet and the interconnection of information systems.

1.2.3 SI objectives

- **A communication tool:**
  - An internal communication tool: IS acts as an intermediary between the SO and the SD.
  - An external communication tool: the IS acts as an intermediary between the company and its environment.
- **A decision-making tool :**
  - The IS provides decision-makers with the information they need to make decisions. It makes it possible to study the foreseeable consequences of decisions and to automate certain decisions.
  - The IS is used to monitor the development of the organisation. It enables internal malfunctions and abnormal situations to be detected.
  - The IS makes it possible to coordinate the activities of the company's various components, particularly those of the SO.

The information system is the memory, the ears, and the word of the organisation.

1.2.4 Qualities of an IS

- Quick and easy access to information: thanks to the use of high-performance software and hardware platforms and user-friendly interfaces.
- Reliability of information: the IS must provide secure, reliable and up-to-date information.
- Relevance of information: information must be filtered according to the user.
- Information security: through the use of effective security software (filtering routers, anti-virus, firewalls, intrusion detectors, etc.).

1.2.5 Components of an IS

The IS varies from one company to another, and may contain all or some of the following elements:

- Company databases ;
- The integrated management software package ;
- The customer relationship management tool ;
- The supply chain management tool ;

- Business applications ;
- Network infrastructure ;
- Data and application servers ;
- Safety devices.

Chapter 2

Decision tables

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2.1 Introduction

In the classical data flow model, every function receives a data flow as input and provides a data flow as output. We now need to describe the internal behaviour of the processing function; this description can take the form of pseudo code, a decision tree or a decision table.

2.2 Definitions

Definition 1:

Decision tables are a representation in the form of a table with several entries. It is a tool for clearly presenting the rules of logic to be used to decide which actions to carry out, taking into account the conditions to be met. They help to clarify texts that are often very difficult to understand and, above all, to detect anomalies and omissions.

Definition 2:

A decision table exhaustively expresses the relationships between the internal logic of a function or process, the data supplied as input and the data returned as output:

— The input data become the conditions of the decision table.

— The output data comes from actions triggered by the decision table.

— The pairs (conditions, actions) become the rules of the decision table or, to put it another way, the internal logic of the function.

The representation of the internal logic of a function by a decision table allows us to achieve the objective of predictability that we expect from a white box (Figure 2.1).

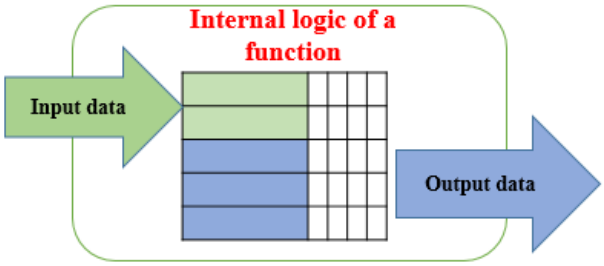


FIGURE 2.1: Definition of a decision table.

2.3 TD structure

A decision table takes the form of a table divided into two parts (2.2):

1. The stump: this is made up of two parts that need to be described:
   — The conditions: these are the proposals to be tested.
   — Actions: to be performed for each set of conditions.
2. The body: this also consists of two parts:
   — Condition values (condition entries): these are the values that a condition can take on

and build a rule.

— Action values (action inputs): these are the actions that must be executed if the corresponding rule is satisfied.

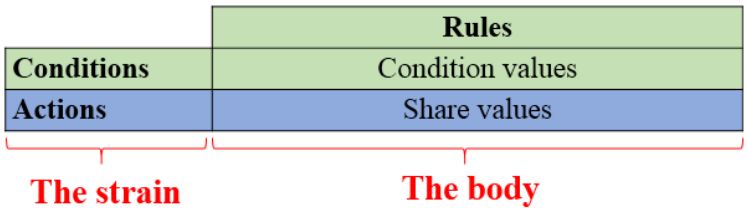


FIGURE 2.2: Structure of a decision table.

2.4 Development of TDs

To draw up the decision table for a process or function, the first step is to identify the conditions and actions.

Example 1:

Let's translate the following text into a decision table: "*Mr X's activities next Friday will depend on two conditions: time and quality of TV programmes. His possible activities will be sport, TV or theatre. He will do sport if the weather is good, whether the TV programmes are interesting or not. He will watch TV if the weather is not fine and if the programmes are interesting. He'll go to the theatre if the weather's not good and if the TV programmes don't interest him*".

Decision table 2.1 shows Mr X's decision logic regarding his Friday activities.

Rules	R_1	R_2	R_3	R_4
Condition 1: Is the weather fine?	O	O	N	N
Condition 2: Are the TV programmes interesting?	O	N	O	N
Action 1: Do the sport	X	X		
Action 2: Watch TV			X	
Action 3 : Go to the theatre				X

TABLE 2.1: Decision table showing Mr X's decision logic regarding his Friday activities.

Example 2: *Let's translate the following text into a decision table: Let X, Y and Z be three real numbers; if they have the same sign, then :* $S = X^2 + Y^2 + Z^2$ *, otherwise:* $S = |X + Y + Z|$.

Rules	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
Condition 1: $X \geq 0$?	O	O	O	O	N	N	N	**N**
Condition 2: $Y \geq 0$?	O	O	N	N	O	O	N	**N**
Condition 3: $Z \geq 0$?	O	N	O	N	O	N	O	**N**
Action 1: $S = X^2 + Y^2 + Z^2$	X							**X**
Action 2: $S = \|X + Y + Z\|$		**X**	**X**	**X**	**X**	**X**	**X**	

2.5 Implementation of TDs

A decision table can be translated as is into executable code, as illustrated by the following pseudo code:

Pseudo code example 1 :

```
If      (the weather is fine) then
        Doing sport
Else
        If (the TV pgs are interesting)then
        Watch TV
        Else
        Going to the theatre
        End
End
```

⌦ In the event of a change within a decision table, adapting the pseudo code can be relatively tedious and, above all, error-prone.

Pseudo code optimisation example 1 :

```
R1 := (the weather is fine).
R2 :=( NOT (the weather is fine)) and (the TV pgs are interesting).
R3 :=( NOT (the weather is fine)) and (NOT(the TV programmes are interesting)).
If      R1 then
        Doing sport
End
If      R2 then
        Watch TV
End
If      R3 then
        Going to the theatre
End
```

2.6 Types of rules

Three types of rule can appear in a decision table: the ELSE rule, simple rules and complex rules.

Rules	R_1	R_2	R_3
Condition 1: Is the weather fine?	O	N	N
Condition 2: Are the TV programmes interesting?	=	O	N
Action 1: Do the sport	X		
Action 2: Watch TV		X	
Action 3 : Go to the theatre			X

TABLE 2.2: Reduction of the decision table in example 1.

The simple rules

A simple rule is a rule that does not have an indifferent value (=) in its condition indicators.

The sign (=) placed in a box means that the condition opposite this box does not apply. Whether the answer is Yes or No, the decision is always executed. The rules R_2 and R_3 in example 1 are simple rules (see table 2.3).

Complex rules

A complex rule is a rule with one or more indifferent values (=) in its condition indicators. R_1 in table 2.3 is a complex rule.

ELSE rule

The ELSE rule covers all the rules that are not in the decision table and generally correspond to errors. In principle, the ELSE rule is placed at the beginning or end of the rules.

Note: The ELSE rule is not taken into account when calculating the number of rules.

2.7 Types of TDs

Three types of table are commonly used: limited tables, extended tables and mixed tables.

2.7.1 Limited tables

A limited table is a decision table where:

— The conditions are indicated in full in the description section. The possible condition indicators are :

  — **Y** for yes ;

  — **N** for no ;

  - for yes or no.

Rules	R_1	R_2	R_3	R_4
C1: Time?	Beau	Beau	Not nice	Not nice
C2: TV programmes?	Interesting	Not interesting	Interesting	Not interesting
Action: Friday activities	Sport	Sport	TV	Theatre

TABLE 2.3: Extended decision table for example 1.

— The actions are indicated in full in the description section. Possible action indicators are :

  — **X** for an action performed ;

  — " " for an action not performed (no mark).

Note: A decision table with only the condition part meeting this criterion is called a limited entry table.

Table 2.1 is a limited decision table.

2.7.2 Extended tables

Extended tables are tables where :

— The condition values represent quantitative details of the conditions.

— The share values represent quantitative details of the conditions.

Note: A decision table with only the condition part meeting this criterion is known as an extended entry table.

2.7.3 Mixed tables

A mixed table is a combination of the limited table and the extended table.

2.8 Calculating the number of rules in a TD

2.8.1 Maximum number of simple rules in a mixed input decision table

The maximum number of simple rules, Nbr_{Max} , that a decision table can contain is calculated by multiplying the number of forms that the different condition entries can take.

$$Nbr_{Max} = F_1 \times F_2 \times F_3 \ldots \times F_n . \quad (2.1)$$

Where F_i is the number of shapes of the inputs for condition i.

— The number of simple rules in the extended inputs corresponds to the number of possible indicators.

— The number of simple rules for limited entries is always equal to 2, i.e. yes or no.

Example:

Rules	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
Condition1: Colour	White	White	Black	Black	Red	Red	Yellow	Yellow
Condition2: availability	O	N	O	N	O	N	O	N

— The extended condition Colour can take 4 values: $_{FColour}$ = {white, *black, red,* yellow}.

— The limited availability condition can take 2 values:

$_{FDavailability}$ = {O, N}.

— The maximum number of simple rules in the table is therefore 8 :

$_{NbMax}$ = $_{Colour\ CF}$ × $_{Availability\ CF}$ = 4 × 2 = 8.

Case of a decision table with limited entries :

Since all the forms of the condition entries in a table with limited entries are 2, the maximum number of simple rules is obtained by raising 2 to the power of the number of conditions in the table.

$$Nbr_{Max} = 2^n . \tag{2.2}$$

Where n is the number of conditions in the table.

Example:

Rules	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
Condition 1: X>=0 ?	O	O	O	O	N	N	N	N
Condition 2: Y>=0?	O	O	N	N	O	O	N	N
Condition 3: Z>=0?	O	N	N	O	O	N	O	N

In this case the number of conditions $n = 3$ so $Nbr_{max} = 2^3 = 8$.

2.8.2 Number of simple rules in a decision table with complex rules

In the presence of complex rules, we must take into account the number of forms of the different conditions for which an indifference indicator appears in order to calculate the number of underlying simple rules, Nbr_{RS} .

$$Nbr_{RS} = RS_1 + RS_2 + RS_3 + \ldots + RS_n \tag{2.3}$$

RS_i is the number of simple rules underlying the RS rule$_i$.

$$RS_i = f_1 \times f_2 \times f_3 \ldots . \times f_n \tag{2.4}$$

f_k is the number of simple values of the input for condition k.

— f_k is 1 for an input with a value.

— f_k equals F_k for an input with an indifference indicator. As a reminder, F_k expresses the number of forms of the inputs for condition k.

Rules	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
Condition 1	B	=	C	B	A	=	D	C
Condition 2	=	O	N	N	N	O	O	N
Condition 3	O	O	O	O	N	N	O	N
Condition 4	1	2	=	3	4	=	4	2
Simple RS equivalent rules	2	4	4	1	1	$4 \times 4 = 16$	1	1

- $FCondition_1 = 4$ because condition 1 can take 4 values from {A, *B, C,* D}.
- $FCondition_2 = FCondition_2 = 2$ because condition 2 and 3 can take 2 values from {O, N}.
- $FCondition_4 = 4$ can take 4 values from {1, 2, 3, 4}.

2.9 Completing a TD

A table is said to be "complete" when all combinations of condition values are covered.

2.9.1 Logically complete table

A table is said to be logically complete if all the missing rules are covered by the ELSE rule. In principle, the ELSE rule leads to error processing.

Example:

Rules	Else	R_1	R_2	R_3	R_4	R_5
Condition 1		O	O	N	N	N
Condition 2		N	N	O	O	N
Condition 3		N	N	N	N	O
Condition 4		O	N	Y N	=	
Simple RS equivalent rules	10	1	1	1	1	2

— Number of effective simple rules= 1+1+1+1+2=6.

— Maximum number of simple rules= $2^n = 2^4 = 16$

— Number of Else rules = 16-6=10. So the table is complete.

2.9.2 Formally complete table

A table is said to be formally complete when no rule is missing and the ELSE rule does not appear.

Example:

Rules	R_1	R_2	R_3	R_4	R_5
Condition 1	O	O	O	O	N
Condition 2	O	O	O	N	=
Condition 3	O	O	N	=	=
Condition 4	O	N	=	=	=
Simple RS equivalent rules	1	1	2	4	8

2.10 Exercises

Exercise 1

Below is a decision table:

	R1	R2	R3	R4	R5
Luxury" version	O	O	O	N	N
Colour	Red	Green	Blue	=	Green
Class	1	3	=	2	4
Packed	=	O	O	=	O
Action 1	X		X		X
Action 2		X		X	

Questions :

1. Calculate the number of equivalent simple rules in this table.
2. How many simple rules can be obtained with this set of conditions (give the calculation formula)?
3. How many simple rules are missing from this table to ensure that it covers all cases?
4. What type of table is this?

Exercise 2

Algerian telephone operator *Mobilis* is launching a new 100% football offer, the *El-Khadra* offer. With the purchase of this SIM card, fans can take advantage of a welcome credit equal to 500 DA, in addition to the advantages below:

— A GOAL for the Greens = 10 free text messages.

— A win for the Greens = 100 DA of free credit.

— Bonus equals 50% on all top-ups up to 500 DA.

Questions :

— Draw up a list of the conditions and actions included in the previous algorithm.

— Calculate the number of rules (give the formula).

— Create a decision table with the appropriate conditions, actions and decision rules

2.11 Conclusion

A decision table offers the following advantages:

— Easy to read, even when there are many conditions, actions and rules ;

— A precise representation of the internal logic of the function ;

— Thanks to the completeness mechanism, it is possible to guarantee the exhaustiveness of the rules applicable to the various data received as input.

Chapter 3

Information coding

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3.1 Introduction

In order to be processed by a computer, information needs to be structured, and this structuring necessarily involves associating codes with the different types of information handled by the information system, so that each piece of information can be clearly and uniquely identified. The information that circulates within an organisation is very important for its smooth running and for the decisions taken by its managers, which is why an error in this information can be fatal to the organisation. To avoid such errors, controls must be in place. The aim of this chapter is to introduce the concepts of coding and data control.

3.2 The raw material of an IS: information

3.2.1 Definitions

Definition 1

The concept of information is becoming ubiquitous in our society, becoming a veritable economic sector, and creating its own constraints. Information is an *abstract* notion, an element in a chain of communication, whatever it may be.

If information is considered independently of a communication condition, it is just data; it is also a form without considering its content. Information is therefore data used by the right person (or entity), at the right time and in the right place.

Definition 2

This concept covers one or more elements of knowledge (of various origins) that can be coded with the aim of being stored, processed or communicated. This transmission of information is therefore likely to trigger an action or process. It then becomes necessary, even imperative, to group together the information, classify it and structure it.

Information is a :

— Complex set of data and links.

— Describes an activity.

— Several possible values.

Information is an informal element, either raw, which we will call data, or which constitutes the processing rules, in fact the programme or software.

3.2.2 Data, information and knowledge

a) **Data :**

— Data is the direct result of a measurement.

— It can be collected by a monitoring tool, by an individual or already present in a database, for example.

— Data alone is not enough to make a decision about what action to take.

b) **Information :**

— Information is data that has been given a meaning and interpretation.

— Information enables an operational manager to make a decision (on a local or small scale) about an action to be taken.

The relationship between data and information is shown in Figure 3.1. The task of IT is to model. In other words, to automatically apply certain rules (simple or complex) to data in order to obtain information and deliver it quickly to decision-makers.

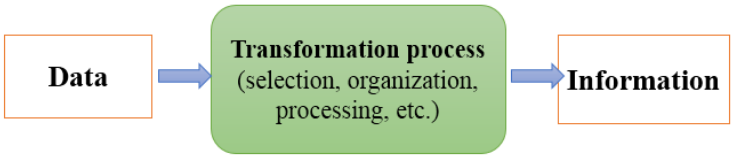


FIGURE 3.1: Relationship between data and information.

c) **Knowledge:** Knowledge is the result of a manager's reflection on analysed information based on :

— its experiences, ideas and the opinions of others.

— its own expertise.

Figure 3.2 describes the relationship between the two concepts: information and knowledge.

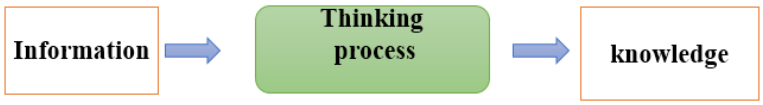


FIGURE 3.2: Relationship between information and knowledge.

The link between the different concepts is summarised in Figure 3.3.

3.3 Data coding

3.3.1 Definition

Coding is an operation that substitutes information in its natural form with an abbreviated symbol (code) that is better adapted to the user's objectives and to automatic processing (Figure 3.4).

A code: is an abbreviated name or a representation of the information used to designate an object or a concept in a clear and unique way.

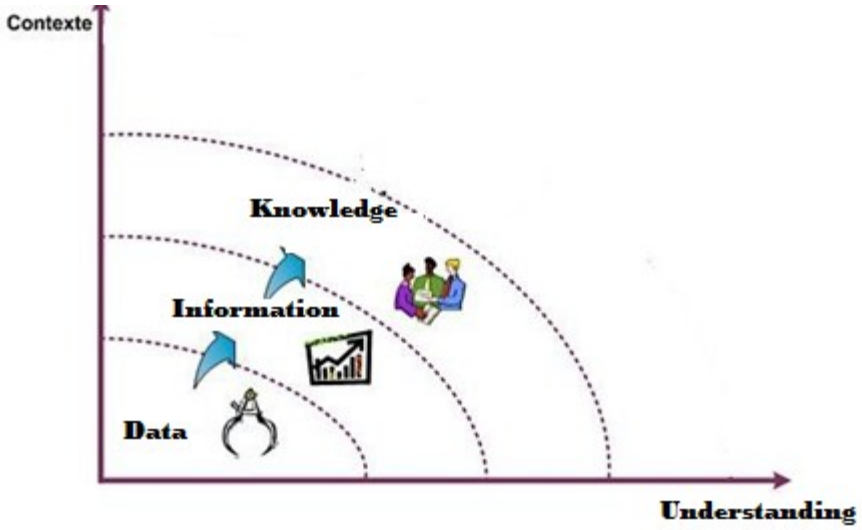


FIGURE 3.3: Relationship between data, information and knowledge.

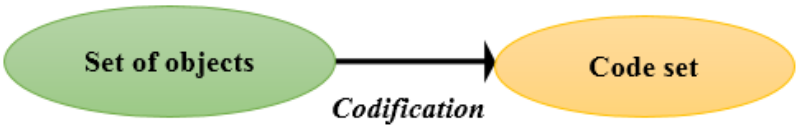


FIGURE 3.4: Definition of coding.

3.3.2 Codification objectives

1. Unambiguously identify an item in a file or an individual in a set.
2. Saves space and time: objects are often coded to avoid having to handle cumbersome names. This saves time during reading and writing operations and space for recording on different media.
3. Represent certain properties of an object: reading the code alone can sometimes identify certain properties of the object. Example: the personnel number of an employee in a company: [date of incorporation][Category][Sequential number]. The code [2009][13][010] tells us that the employee joined in 2009, is in category 13, is an engineer and is the tenth engineer.

3.3.3 Coding criteria

The coding must meet a minimum number of criteria:

1. Unambiguous identification :
   - Uniqueness: a code must designate one and only one object.
   - Each object has its own code (see Figure 3.5).

2. Durability: a code must be durable. In particular, it must always allow for extensions and insertions, as the set of objects may increase over time.
   - — Extension means that the totality of codified information can be increased. Insertion means that a new code can be inserted between two existing codes.
   - — The codification must allow for the extension and insertion of new objects without calling into question the chosen codification, i.e. the codification established must be as stable as possible.
3. Conciseness: reflects the fact that a code must be clear and short, without neglecting the possibility of the evolution of all the information to be codified.

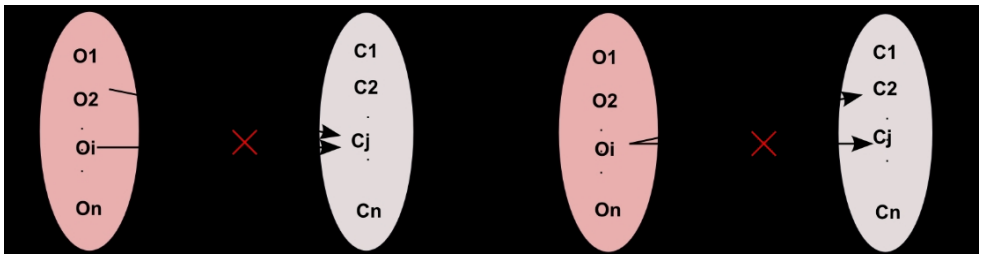


FIGURE 3.5: Unambiguous identification of objects in a set.

3.3.4 Types of coding

3.3.4.1 Sequential coding

This involves assigning consecutive numbers to the objects to be coded in the same set. Example: the employees of a company are coded in the chronological order in which they were hired.

Advantages :

- Unambiguously identify objects.
- Simple object coding.
- Possibility of extensions.

Disadvantages

- Impossible to insert.
- It is not significant.

3.3.4.2 Coding by band

It consists of dividing all the objects to be coded into several categories, and each category of objects is allocated a code band. Within the bands, the codes are generally sequential. Example: In the library, books are classified by category:

— Technology : [001 to 100]
— Literature [101 to 200]
— Sociology [201 to 300]
— Medicine [301 to 400]
— General culture [401 to 500]

Advantages :

- Unambiguously identify objects.
- Simple object coding.
- Possibility of extensions (adding a new tranche).

Disadvantages

- Impossible to insert.
- It is not significant.
- Number of codes in a bracket that is difficult to set.
- The division of objects into categories is not always obvious.

3.3.4.3 Articulated coding (juxtaposed)

The codes are divided into several zones (fields), each with a specific meaning.
Example 1: Personnel number of an employee in a company [99][99][999] Where :

— 99: year of entry.
— 99: professional category
— 999: sequential number

Example 2: Vehicle registration Figure 3.6 shows an example of an articulated code.

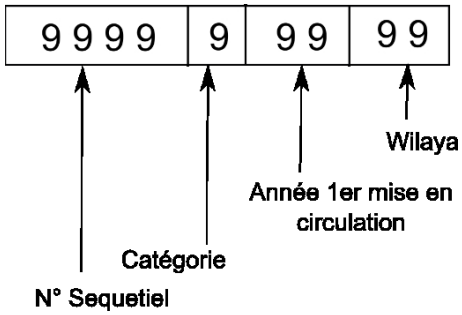


FIGURE 3.6: Car coding.

Advantages :

- Widely used
- Unambiguously identify objects.

- Integration and extension possible.
- Possibility of grouping objects according to a given criterion.
- Possibility to control.

Disadvantages

- Code is too long and therefore difficult to manipulate

3.3.4.4 Mnemonic code

It consists of abbreviating the designation of objects using a reduced set of characters that is evocative of the codified object. Example: Invoice → fact. Receipt slip → rec. slip

Advantages :

- Easy to install.
- Significant.

Disadvantages

- Relates to the name, not the value
- Used only to code variables in a computer program.

3.3.5 Case study : Product coding (barcodes)

3.3.5.1 Definition

Barcodes are a discreet part of everyday life. Designed to automate the acquisition of generally digital information, they are used in fields as varied as library loan management, optical cash registers and production control in industry...

All information can be encoded in the form of binary words, which is the basis of digital electronics. Various processes have been developed to enable fast, automatic data acquisition. Bar codes are particularly well suited to encoding a numeric or alphanumeric string of no more than a few dozen characters.

Figure 3.7 shows an example of a barcode.



FIGURE 3.7: Example of a barcode.

3.3.5.2 History

The first step towards today's barcodes was taken in 1948, when Bernard Silver, a graduate student, overheard a conversation in the lobby of Philadelphia's Drexel Institute of Technology.

The president of a chain of food shops asked one of the deans to research a means of automatically capturing data on products passing through the checkout. The dean ignored the request, but Bernard Silver reported the conversation to his friend Norman Joseph Woodland, a 27-year-old graduate student and professor at Drexel. The problem fascinated Woodland.

3.3.5.3 Reading a code

There are two common ways of reading barcodes:

a) **Optical reading:** This involves shining a beam of light (often a very low-power laser) onto the barcode and then analysing the reflected light. Optical readers such as *handheld scanners* can now be connected to all microcomputers.

b) **Magnetic reading:** The bar codes used on cheque books are printed using magnetisable ink. The principle is the same as that of a magnetic stripe reader: a read head translates the succession of magnetised bars into electrical signals.

3.3.5.4 Different types of barcode

a) **Linear barcodes**



FIGURE 3.8: Linear barcodes.

b) **2D barcodes**

3.3.5.5 EAN standardisation

— The EAN (*European Article Number*) is a standard guaranteeing that an item's barcode will be recognised in all European Union countries.

— The EAN also ensures compatibility with the U.P.C. codes used in North America.

— The EAN barcode: An international standard for coding everyday consumer items, the barcode generally consists of 13 digits (EAN13). It can be reduced to 8 digits (EAN8) for small items where it is not possible to fit the EAN13 barcode.



FIGURE 3.9: 2D barcodes.

3.3.5.6 Composition of the EAN-13 bar code

The code word consists of 13 characters divided between three zones (Figure 4.2). Reading from left to right, we find :

— The *flag* encodes the origin of the code in two or three characters.

— The *useful coding zone*, comprising 10 characters, the first 4 or 3 of which represent the manufacturer and the last 6 the item (in the case of a 13-digit code). In the case of this zone, the coding depends on the original zone.

— The last character is the *control code* that determines the validity of the code. It is calculated using a standard algorithm.

Note: In view of the large number of countries using EAN-13 barcodes, it was decided that the country code could be either 2 or 3 characters long. In the latter case, the manufacturer code is only 3 characters long.

3.3.5.7 Interpretation of the code according to the EAN standard

The different characters are represented by the concatenation of two basic symbols of widths equal to 0.5 mm. There are two basic symbols:

— the black bar, which represents the binary 1,

— the white bar, which represents the binary 0.

- The code must be readable from left to right and right to left, so it has been divided into two parts by a separator.

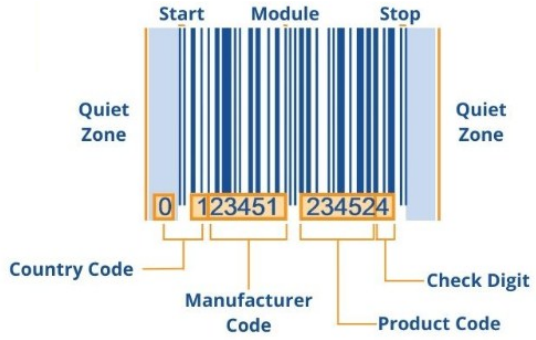


FIGURE 3.10: Composition of the EAN-13 barcode.

- **01010 is** found between the right and left parts of the code and **101** at the ends of the code. Note that the separators have bars that are higher than the bars they enclose.
- Each character is encoded using 7 bars, so it is a 7-bit code:

  Three different sets are used depending on the position of the digit in the code. Sets A and B are used only to the left of the centre of the barcode, in no particular order, while set C is used only to the right of the centre of the barcode (Figure 3.11).

Note: The codes in sets A and B all start with 0 and end with 1. Conversely, the codes in set C all start with 1 and end with 0. The coding of a character therefore involves only 5 bits out of 7.

Figure 3.12 shows an example of an EAN-13 code interpreted in binary.

Control character calculation: Here is an example of how to calculate the control character of an EAN-13 code.

EAN Code: 4007630 **00011X**

EAN	4	0	0	7	6	3	0	0	0	0	1	1
Multiplication	1	3	1	3	1	3	1	3	1	3	1	3
Results	4	0	0	21	6	9	0	0	0	0	1	3
Sum	4+	0 +	0 +	21 +	6 +	9 +	0 +	0 +	0 +	0 +	1 +	3 = 44
Division by 10	44 / 10 =4 ➔ remain=4											
Control character	10-remain =6											

Note: If the remainder of the division is 0, then the control character is also 0.

The final EAN code is therefore 4 007630 000116. This is shown in Figure 3.13.

Caractère codé	Jeu A	Jeu B	Jeu C
0	0001101	0100111	1110010
1	0011001	0110011	1100110
2	0010011	0011011	1101100
3	0111101	0100001	1000010
4	0100011	0011101	1011100
5	0110001	0111001	1001110
6	0101111	0000101	1010000
7	0111011	0010001	1000100
8	0110111	0001001	1001000
9	0001011	0010111	1110100

FIGURE 3.11: 7-bit coding for the EAN-13 barcode.

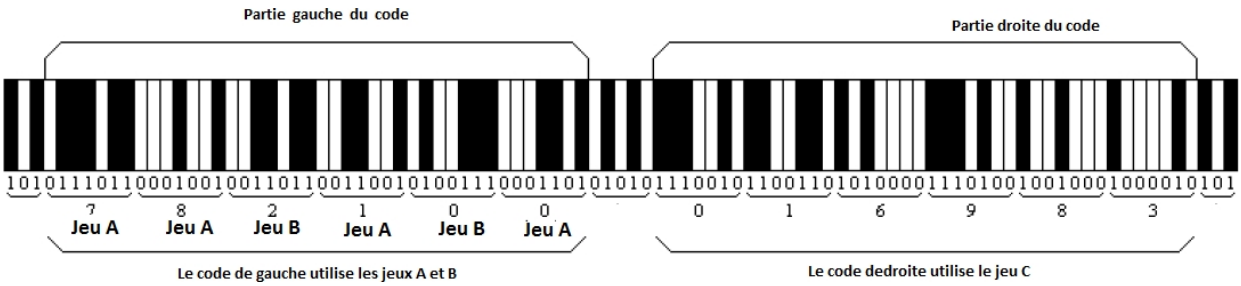


FIGURE 3.12: 7-bit coding for the EAN-13 barcode.



FIGURE 3.13: Example of an EAN-13 barcode.

3.4 Exercise

The National Student Code (CNE) is used to assign a code to baccalauréat candidates. The CNE is a unique number given to each baccalauréat candidate.

Notes: The Algerian lycée education system has five streams: Mathematics, Natural Science, Management, Literature and Foreign Languages, Literature and Philosophy. The number of students per stream does not exceed 10,000 (per wilaya).

Questions :

1. Propose a useful and meaningful CNE code.
2. Propose a sequential coding system.
3. What is the disadvantage of this codification?
4. If you opt for coding by band :
   a) Suggest two different possibilities
   b) For each option, indicate the number of slices and the slice size.